

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE
NAME

CLASS

INDEX NUMBER

CHEMISTRY

8872/02

Section A: Structured Questions

18 September 2013

Section B: Free Response Questions

2 hours

Candidates answer Section A on the Question Paper

Additional Materials: Writing Papers

Data Booklet

Cover Page

READ THESE INSTRUCTIONS FIRST

Write your index number, name and civics group.

Write in dark blue or black pen.

You may use pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the space provided.

A Data Booklet is provided.

You are advised to show all working in calculations.

You are reminded of the need for good English and clear presentation in your answers.

You are reminded of the need for good handwriting.

Your final answers should be in 3 significant figures.

You may use a calculator.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	8
2	9
3	8
4	15
Significant figures	
Handwriting	
Total	40

This document consists of **17** printed pages and **1** blank page.

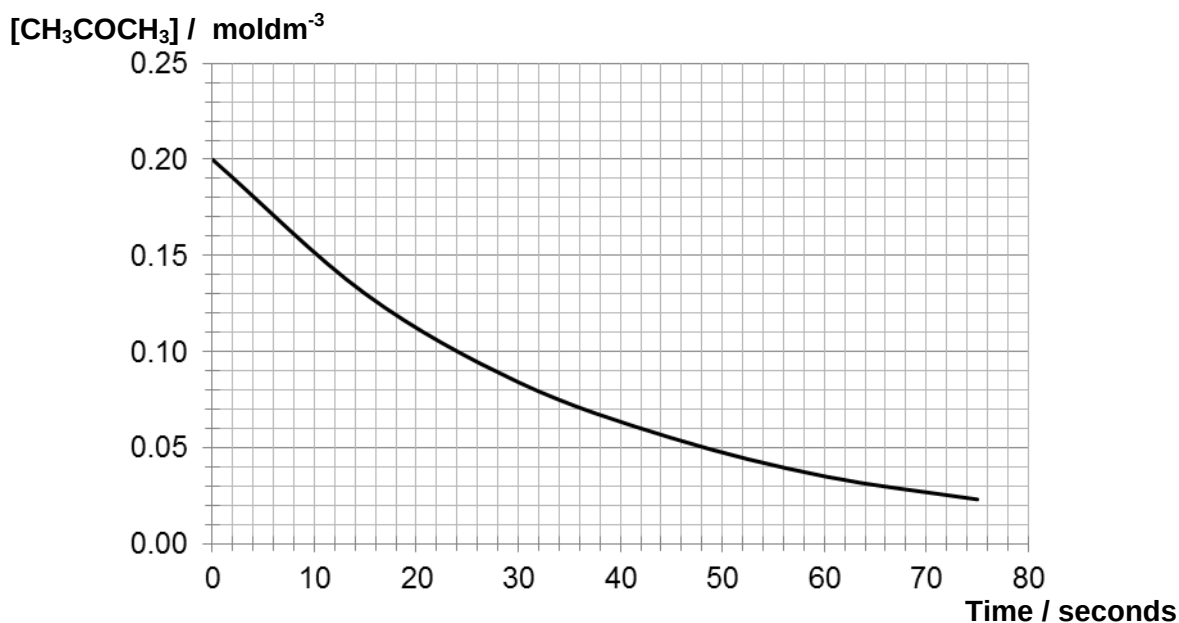
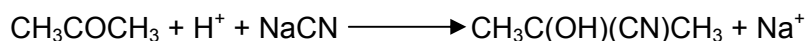


Section A

Answer **ALL** questions on the spaces provided.

- 1 The kinetics of the reaction between acetone and potassium cyanide was investigated.

An experiment was performed in which $0.200 \text{ mol dm}^{-3}$ of acetone, CH_3COCH_3 was reacted with an excess of acid and sodium cyanide, NaCN . A graph of concentration of CH_3COCH_3 against time was plotted.



- (i) Using your graph, deduce the order of reaction with respect to acetone.

Workings of two $t_{1/2}$ shown on graph

$t_{1/2}$ is approximately constant at 24 s

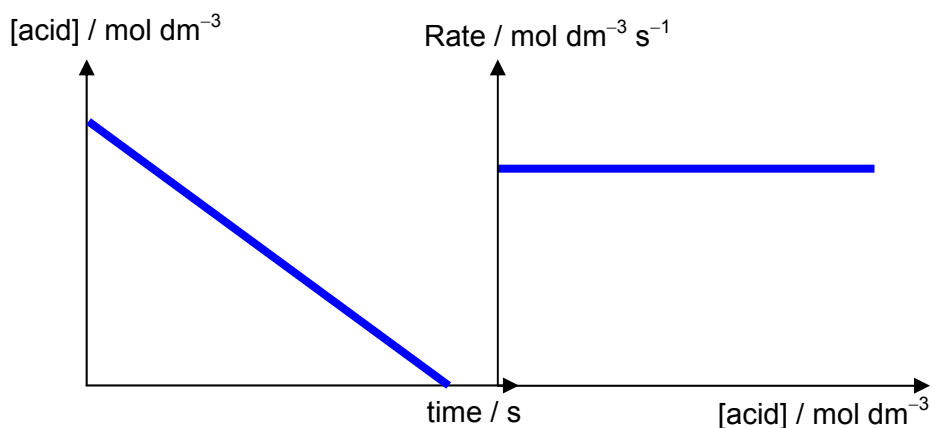
⇒ order of reaction w.r.t. propanone is 1

- (ii) Why was the acid and sodium cyanide used in excess?

To ensure their concentration remain relatively constant during the reaction and not affect the rate of reaction

- (iii) The order of reaction with respect to the acid was zero.

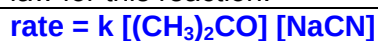
Sketch, on the axes provided, the shape of the graphs relating to the information provided.



Graph 1

Graph 2

- (iv) Given that the order of reaction with respect to sodium cyanide is one, write the rate law for this reaction.



- (v) Suggest with a reason whether pentan-2-one reacts more or less rapidly with NaCN as compared to acetone.

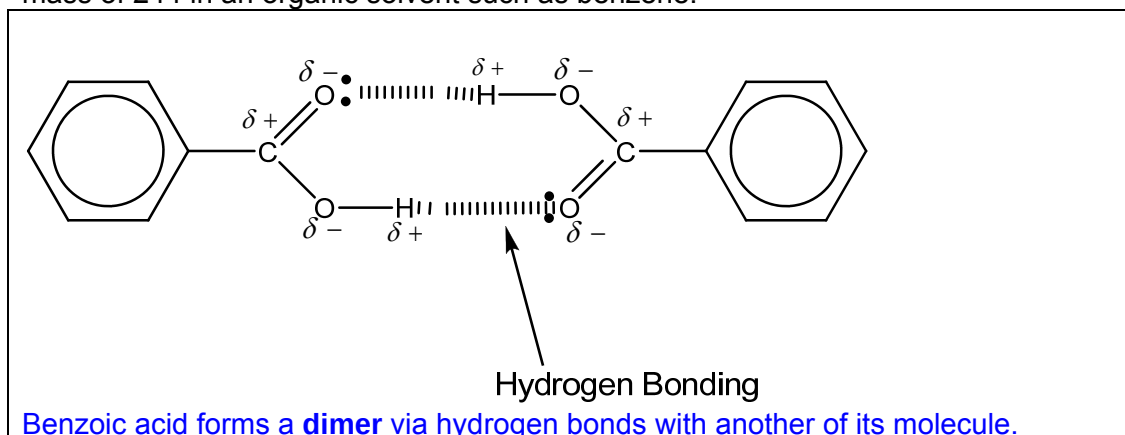
Less rapidly

There is a bulkier propyl group/ $\text{CH}_3\text{CH}_2\text{CH}_2$ group in pentan-2-one / there is steric hindrance present in pentan-2-one

[Total: 8]

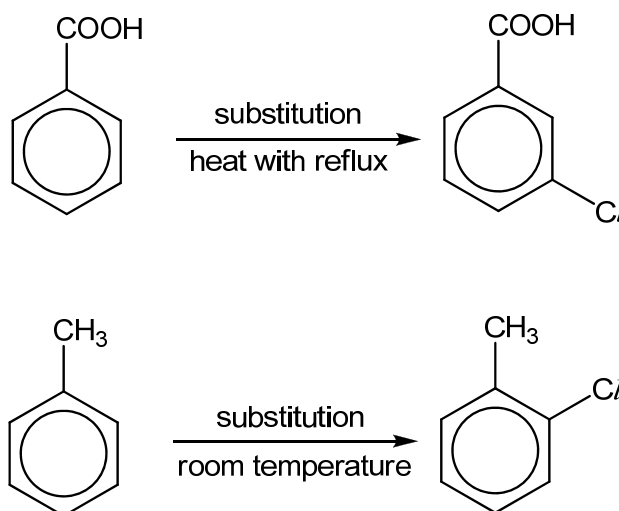
- 2 Benzoic acid is a colourless crystalline solid that has antifungal abilities and is an important precursor for the synthesis of many organic compounds.

- (a) Explain, with the aid of a diagram, why benzoic acid has an apparent molecular mass of 244 in an organic solvent such as benzene.



[2]

- (b) Both benzoic acid and methylbenzene undergo aromatic substitution to produce the chloro-compounds shown:



- (i) Suggest why a harsher condition is required for the substitution of benzoic acid.

–COOH is a deactivating group / electron-withdrawing group and hence makes the benzene ring less reactive towards substitution reactions/ reduces the electron density of the ring

- (ii) How would you expect the acidity of 3-chlorobenzoic acid to compare with that of benzoic acid?

3-chlorobenzoic acid is a stronger acid than benzoic acid. The –Cl atom is an electron withdrawing group and hence will disperse the negative charge on the O atom of 3-chlorobenzoate ion. Thus stabilising 3-chlorobenzoate ion.

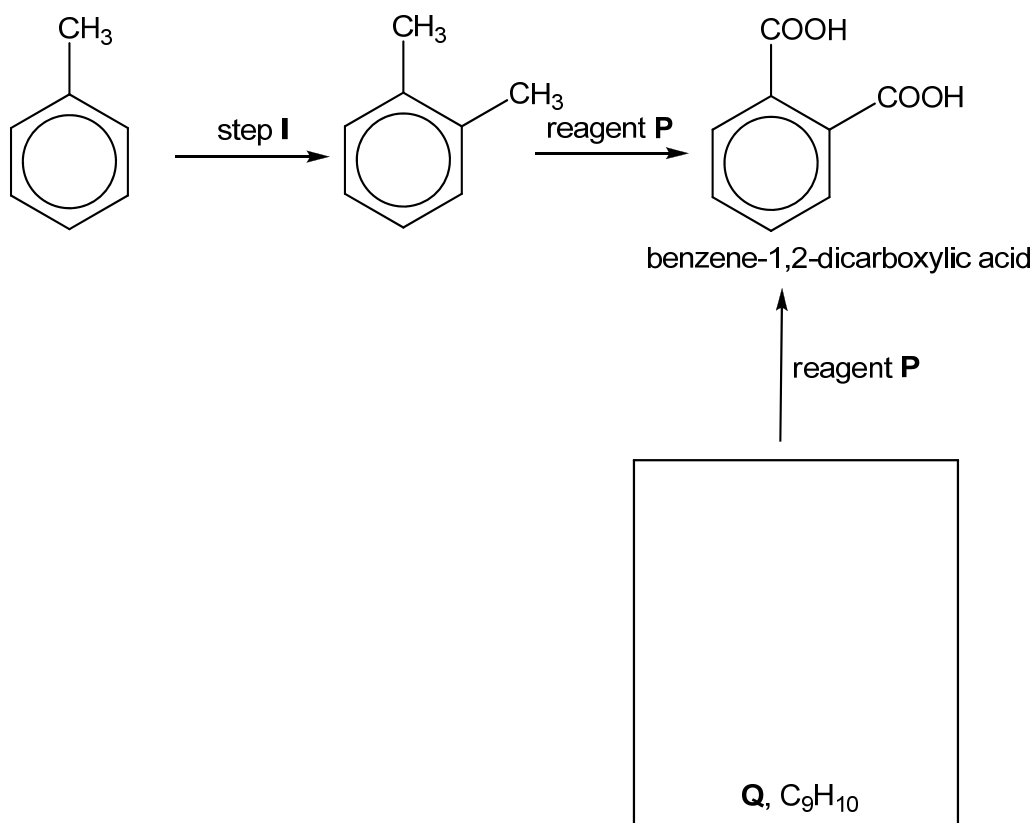
[3]

- (c) Benzoic acid is sparingly soluble in water. However, it dissolves readily to give a colourless solution upon the addition of a reagent, **N**. Identify reagent **N**.

Reagent **N**: **NaOH(aq) or Na₂CO₃(aq)**

[1]

(d) Benzene-1,2-dicarboxylic acid can be produced via the following synthetic pathway.



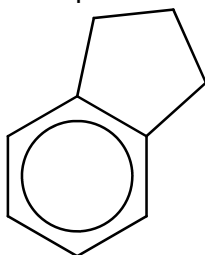
(i) What type of reaction is step I?

Substitution

(ii) Identify reagent **P**.

KMnO₄, dilute H₂SO₄, (heat with reflux)

(iii) Suggest a structure for compound **Q** and draw its structural formula in the box provided.



[3]

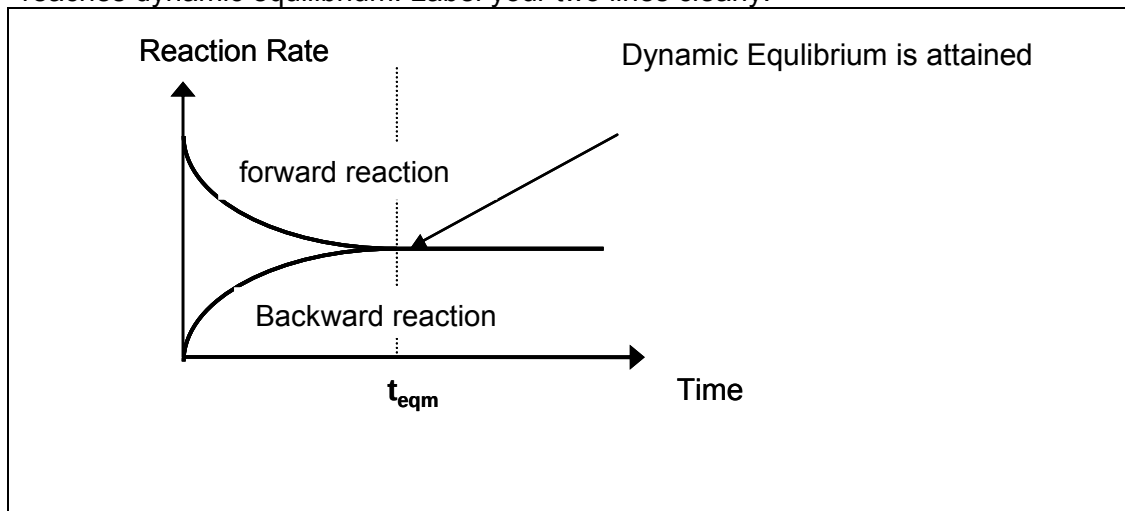
[Total: 9]

- 3 Phosphorus(V) chloride, PCl_5 , is a white solid which sublimes at 160°C .

When gaseous phosphorus(V) chloride is heated in a closed container, the following equilibrium is established.



- (a) Sketch a graph to show how the rates of the forward and backward reactions change from the time phosphorus pentachloride is heated to the time the reaction reaches dynamic equilibrium. Label your two lines clearly.



[2]

- (b) In an experiment, 1.00 mol of PCl_5 vapour was heated in a closed 5.00 dm^3 flask at 500K until equilibrium had been established. The amount of chlorine collected was 0.508 mol.

Write an expression for the equilibrium constant and calculate its value.

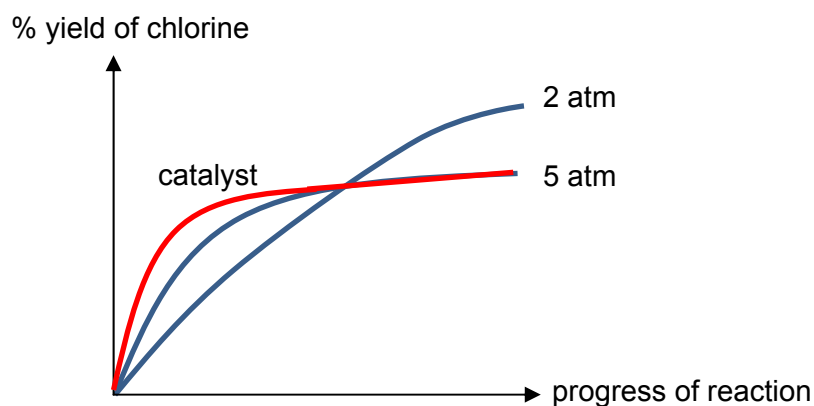
$$K_c = \frac{[\text{PCl}_3][\text{Cl}_2]}{[\text{PCl}_5]} \text{ mol dm}^{-3}$$

	$\text{PCl}_5 \rightleftharpoons$	$\text{PCl}_3 +$	Cl_2
Initial amount/ mol	1.00	0	0
Change in amount/ mol	-0.508	+0.508	+0.508
Eqm amount/ mol	0.492	0.508	0.508

$$K_c = \frac{\left(\frac{0.508}{5}\right)\left(\frac{0.508}{5}\right)}{\left(\frac{0.492}{5}\right)} = 0.105 \text{ mol dm}^{-3}$$

[3]

- (c) The graph below shows how the percentage yield of chlorine from the above equilibrium varies as the reaction progresses at two different pressures.



- (i) Account for the **differences** in the two graphs at different pressures.

At higher pressure (5atm), system responds to decrease the pressure by shifting position of equilibrium to the left to favour the formation of less gaseous molecules. Thus, equilibrium yield of chlorine decreases.

Rate increases at higher pressure, thus gradient becomes steeper.

OR

At lower pressure (2atm), system responds to increase the pressure by shifting position of equilibrium to the right to favour the formation of more gaseous molecules. Thus, equilibrium yield of chlorine increases.

Rate decrease at lower pressure, thus gradient becomes gentler.

- (ii) On the same axes, sketch a graph to show how the percentage yield of chlorine will vary at 5 atm in the presence of a catalyst.

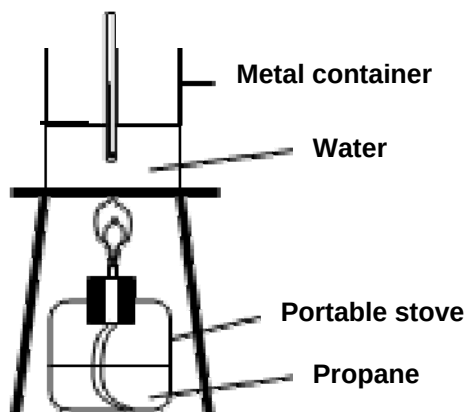
Steeper gradient , same equilibrium yield

[3]

[Total: 8]

- 4 (a) During the last Olympic Games, a torch relay around several countries took place before the torch was used to ignite the Olympic cauldron to mark the start of the Game. The fuel used to light the Olympic flame is propane. Propane is also used in blow torches, portable stoves and in outdoor heaters.

A propane portable stove was used during a barbecue to boil water for making of soup. It took 10 minutes to bring the water to boil.



The following information was known.

Mass of metal container / g	850
Mass of metal container containing water / g	1600
Initial temperature of water / °C	32
Final temperature of water / °C	100
Specific heat capacity of water / J g ⁻¹ K ⁻¹	4.18

Besides the above information, it was also known that the theoretical standard enthalpy change of combustion of propane is $-2220 \text{ kJ mol}^{-1}$.

- (i) Define the *standard enthalpy change of combustion of propane*.

The standard enthalpy change of combustion of propane is the enthalpy change (heat evolved) when 1 mole of propane is completely burnt in oxygen / burnt in excess oxygen under standard conditions of 298K and 1 atm.

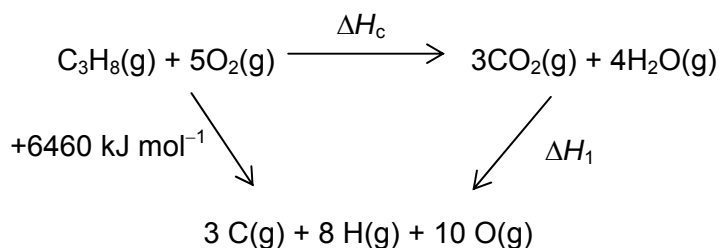
- (ii) Calculate the mass of propane needed to raise the temperature of the water during the barbecue.

Amount of heat gained by water
 $= (1600 - 850) \times 4.18 \times (100 - 32)$
 $= 2.1318 \times 10^5 \text{ J}$
 $= 213 \text{ kJ}$

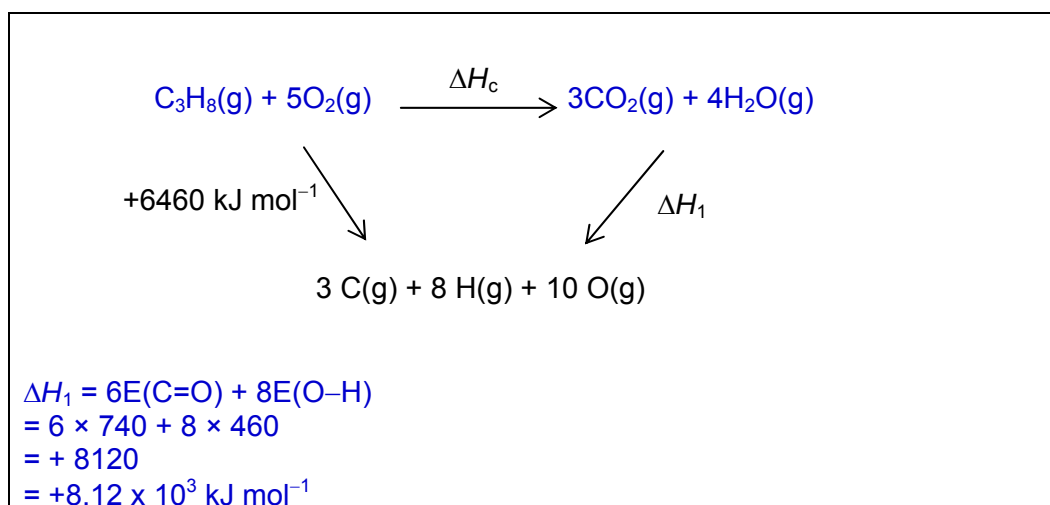
Amount of propane needed $= 213.18 \div 2220 = 0.09603 \text{ mol}$

Mass of propane required $= 0.09603 \times 44.0 = 4.225 = 4.23 \text{ g}$

Another value of the standard enthalpy change of combustion of propane, ΔH_c , can be calculated using bond energies and the energy cycle below.



(iii) Use suitable bond energies given in the *Data Booklet* to calculate ΔH_1 .



(iv) Hence, calculate ΔH_c .

$$\begin{aligned}
 \Delta H_c &= +6460 - (+8120) \\
 &= -1660 \\
 &= -1.66 \times 10^3 \text{ kJ mol}^{-1}
 \end{aligned}$$

(v) Suggest why the value of ΔH_c you calculated differs from the theoretical value.

The bond energy values obtained from the *Data Booklet* are average (mean) bond energy values derived from a full range of molecules that contain the particular bond and would differ from the experimental values.

[7]

- (b) Most outdoor heaters are powered by the combustion of propane gas stored in small cylinders. At room temperature and pressure, a typical outdoor heater designed to produce 15 kW of energy runs from a cylinder containing 13 kg of propane.

As pure propane gas is odourless, small amounts of another compound are usually added so that gas leaks from propane cylinders can be detected. An example of such an odorant is ethanethiol, $\text{C}_2\text{H}_5\text{SH}$ which has an odour that resembles that of onion. Ethanethiol is chosen since the human nose can detect its presence at levels of only about 0.02 moles of it per billion (10^9) moles of propane.

- (i) Calculate the total amount of heat energy released by the complete combustion of all the propane in a cylinder used in an outdoor heater.

$$\begin{aligned}\text{Total amount of heat energy released} &= \frac{13000}{44.0} \times 2220 \\ &= 6.55 \times 10^5 \text{ kJ}\end{aligned}$$

- (ii) Using your answer to (b)(i), calculate the time needed to empty one cylinder of propane to produce 15 kW of energy. You are given that 15kW is equivalent to 15 kJ s^{-1} .

$$\begin{aligned}\text{Total amount of time needed to empty one cylinder} &= \frac{6.55 \times 10^5}{15} \\ &= 4.37 \times 10^4 \text{ s}\end{aligned}$$

- (iii) Hence, calculate the rate, in $\text{cm}^3 \text{ s}^{-1}$, at which propane must leave the cylinder in order to produce 15 kW of energy.

$$\begin{aligned}\text{Rate at which propane must leave the container} &= \frac{\frac{13000}{44.0} \times 24000}{4.37 \times 10^4} \\ &= 162 \text{ cm}^3 \text{ s}^{-1}\end{aligned}$$

- (iv) Calculate the volume of carbon dioxide evolved when 13 kg of propane was completely burnt in an outdoor heater.

$$\begin{aligned}\text{C}_3\text{H}_8(\text{g}) + 5\text{O}_2(\text{g}) &\rightarrow 3\text{CO}_2(\text{g}) + 4\text{H}_2\text{O}(\text{g}) \\ \text{Amount of propane burnt} &= 13000 \div 44 = 295.45 \text{ mol} \\ \text{Amount of CO}_2 \text{ evolved} &= 3 \times 295.45 = 886.36 \text{ mol} \\ \text{Volume of CO}_2 \text{ evolved} &= 886.36 \times 24 = 21273 \text{ dm}^3 \\ &= 2.13 \times 10^4 \text{ dm}^3\end{aligned}$$

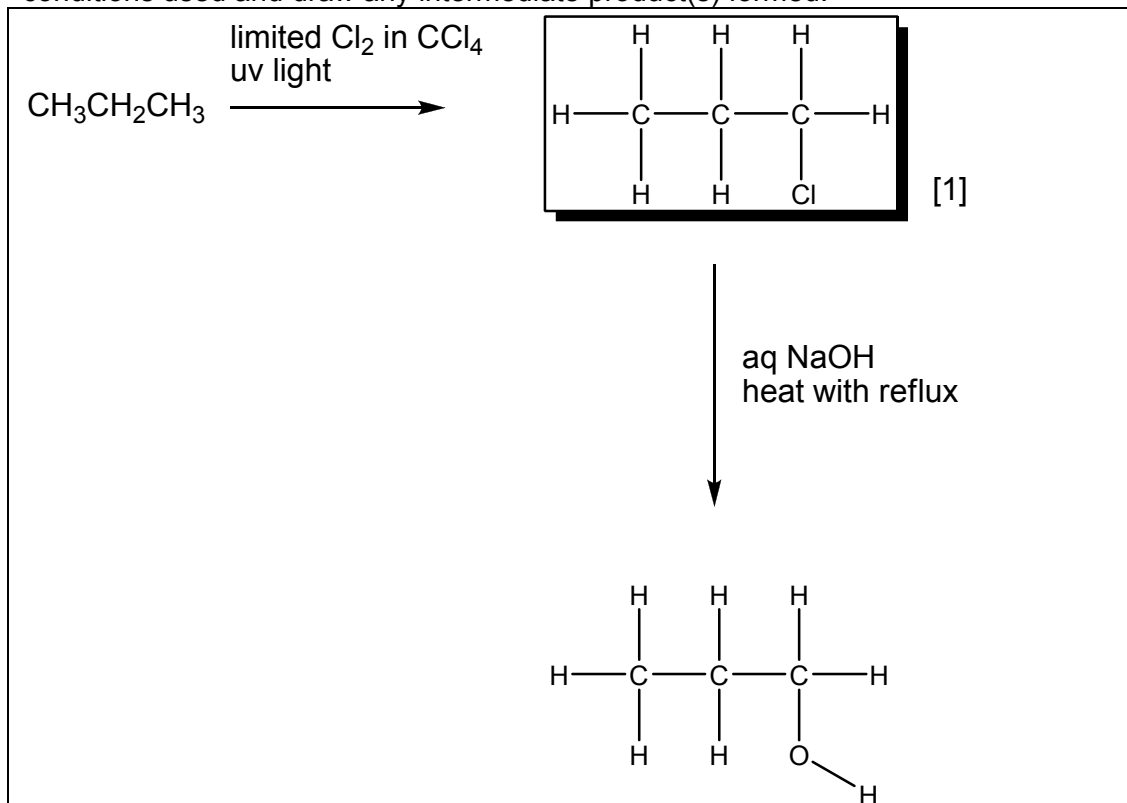
- (v) Calculate the minimum mass of ethanethiol which must be added to 13 kg of propane so that it can be detected by the human nose in the event of a propane gas leak.

$$\begin{aligned}\text{Minimum mass of ethanethiol required} &= \frac{\frac{x}{62.1}}{\frac{13000}{44.0}} \times 10^9 = 0.02 \\ \text{Solve for } x &= 3.67 \times 10^{-7} \text{ g}\end{aligned}$$

[5]

- (c) Besides being used as fuel, propane can also be used to synthesise other organic compounds.

Outline the synthesis of propan-1-ol from propane. State clearly the reagents and conditions used and draw any intermediate product(s) formed.



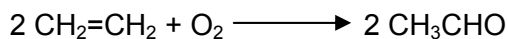
[3]

[Total: 15]

Section B

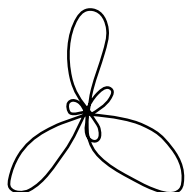
Answer **two** of the three questions in this section on separate answer paper.

- 5 (a) Ethanal is one of the most important aldehydes occurring widely in nature and is also the cause of hangovers from alcohol consumption. It is produced on a large scale industrially via the Wacker Process as shown.



- (i) State the hybridisation of each carbon atom in ethene and draw a diagram to show the hybridised orbitals around one of these carbon atoms.

Sp² hybridisation



- (ii) Describe, in terms of orbital overlap, the bonding of the two carbon atoms of the C=C bond in ethene.

The two carbon atoms are sp² hybridised. One sp² orbital from each carbon atom overlap head-on to give a sigma bond. The unhybridised p orbital from each carbon atom overlap sideways to give π bond.

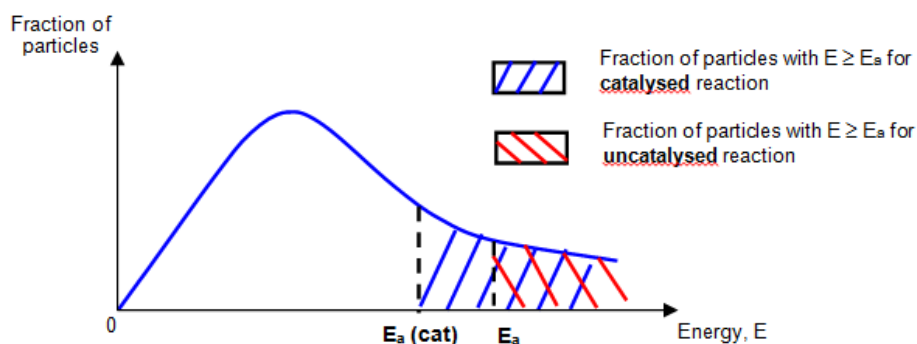
[4]

- (b) The Wacker reaction is accelerated by palladium catalysts.

- (i) Define *catalyst*.

A catalyst is defined as a substance which alters the rate of reaction by providing an alternative pathway with a lower activation energy and itself remains chemically unchanged at the end of the reaction.

- (ii) Explain with the aid of a diagram that shows the distribution of molecular energies, how the addition of a catalyst accelerates the reaction.



- Catalyst provides an alternative pathway with a lower activation energy.
- Fraction of particles with energy equal to or greater than the activation energy of the catalysed reaction, $E_{a(\text{cat})}$ increases.
- Frequency of effective collisions taking place in the reaction increases.
- Since rate of reaction is proportional to the frequency of effective collisions, rate of reaction increases.

- (iii) In a college laboratory, ethanal can be synthesised from ethanol. Briefly outline how you would carry out this reaction.

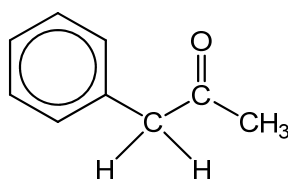
Ethanal can be prepared by heating ethanol under distillation with potassium dichromate and dilute sulfuric acid.

- (iv) Explain why ethanal can be prepared in the way you described in (b)(iii), in terms of the volatility of ethanal and ethanol.

[8]

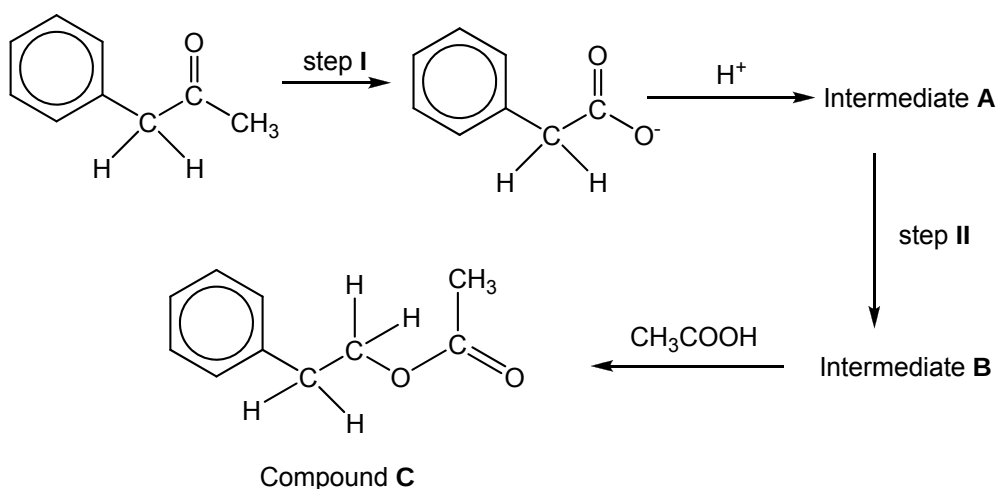
- Both ethanal and ethanol have simple molecular structures.
- There exists hydrogen bonding between ethanol molecules but van der waals forces of attraction between ethanal molecules.
- Less energy is needed to overcome the weaker van der waals forces of attraction between ethanal molecules.
- Hence volatility of ethanal is higher and will distil out before it can continue to get oxidised to ethanoic acid.

- (c) (i) Another common carbonyl compound used in the industry is phenylacetone, which is an intermediate used to produce pesticides and anticoagulants.



phenylacetone

Compound **C** can be synthesised from phenylacetone.

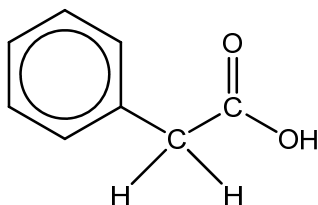


Complete the synthetic route by giving the structures of intermediates **A** and **B** as well as the reagents and conditions for steps I and II.

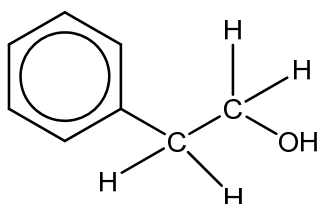
step I
I₂ (aq), NaOH(aq), warm

step II
LiAlH₄, dry ether, room temperature

Intermediate A

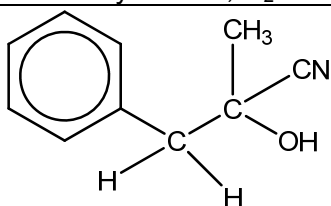


Intermediate B

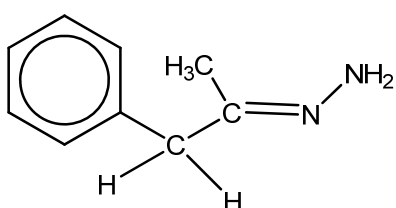


(ii) Suggest the structures of the product and type of reaction undergone when phenylacetone reacts with the following reagents.

- HCN, trace OH^-
- hydrazine, $\text{H}_2\text{N}-\text{NH}_2$



Type of reaction: Addition



Type of reaction: condensation

[8]

[Total: 20]

- 6 (a) The elements of the third period of the Periodic Table, sodium to sulfur, all form chlorides by direct combination.

- (i) When dry chlorine is passed over heated aluminium powder in a long hard-glass tube, a vapour is produced which condenses to a solid in the cooler parts of the tube. At low temperatures, the vapour has a M_r of 267.

State one observation you would make during this reaction.

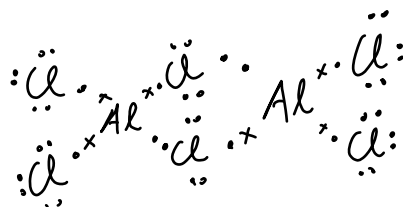
Any one observations:

- (yellow) White solid, AlCl_3 formed/ white fumes, Al_2Cl_6 seen
- Chlorine gas decolourised
- Aluminium glows

- (ii) Suggest the molecular formula of the vapour, and draw a dot-and-cross diagram to illustrate its bonding.

MF of vapour: Al_2Cl_6

Al_2Cl_6 is formed through two dative covalent bonding

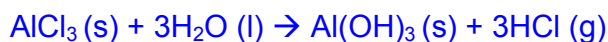


- (iii) The solid reacts with water in two separate ways.

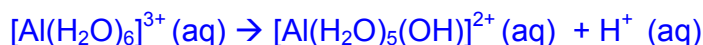
- When a few drops of water are added to the solid, steamy white fumes are evolved and a white solid remains, which is insoluble in water.
- When a large amount of water is added to the solid, a clear, weakly acidic solution results.

Write equations, including state symbols, for these two reactions.

Few drops of water:



Excess water:



OR

The 2 equations can be combined:



[7]

- (b) Lattice energy can be used as a measure of the energetic stability of ionic compounds.

(i) Define the *lattice energy of silver fluoride*.

Lattice energy of silver fluoride is the enthalpy change which occurs when 1 mole of solid ionic silver fluoride is formed from its constituent gaseous ions ($\text{Ag}^+(\text{g})$ & $\text{F}^-(\text{g})$) under standard conditions of 298K, 1 atm.

The table below shows numerical values of lattice energies for silver fluoride and silver iodide. These have been determined from experimental data or theoretically calculated.

Compound	Experimental value / kJ mol^{-1}	Theoretical value / kJ mol^{-1}
AgF	−967	−953
AgI	−889	−808

(ii) By quoting appropriate data from the Data Booklet, explain why the lattice energy of silver iodide is less exothermic than silver fluoride.

- Lattice energy $\propto \frac{q_+ \times q_-}{r_+ + r_-}$
- F^- and I^- have the same charges
- but **ionic radius** of F^- is smaller than I^-
- quote ionic radius of both F^- (0.136nm) and I^- (0.216nm)
- lattice energy of AgI is less exothermic than AgF

(iii) Silver fluoride and silver iodide have the same crystal structure. There is close agreement between the experimental and theoretical values of lattice energy for AgF but not for AgI. Suggest a reason for this.

AgI is an ionic compound with some covalent character due to large electron cloud of I^- that can be polarized by small size Ag^+ .

[4]

- (c) Compound **C** has the empirical formula CH_2O and M_r of 90. There is no reaction when **C** is treated with NaHCO_3 . When 0.600 g of **C** is reacted with an excess of Na, 160 cm^3 of H_2 , measured at room temperature and pressure, is produced. Treatment of **C** with 2,4-dinitrophenylhydrazine reagent produces an orange solid. When **C** is warmed with Fehling's reagent, a brick red precipitate is formed.

The structure of **C** has the following feature:

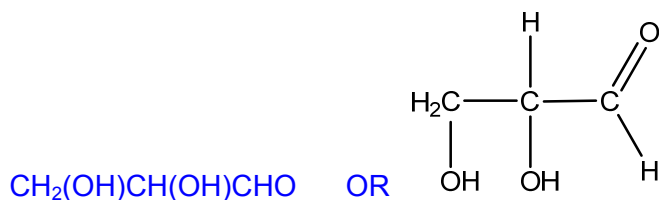
- No carbon atom has more than one oxygen atom joined to it.

Suggest a structure for **C**, explaining your reasoning.

- ✓ Since **C** has no reaction with $\text{NaHCO}_3 \Rightarrow$ carboxylic acid absent
- ✓ Since **C** undergoes redox/acid-metal reaction with Na $\Rightarrow$
- ✓ alcohol present
- ✓ $n(\text{H}_2)$ produced from 0.600 g of **C** = $\frac{160}{24000} = 6.67 \times 10^{-3} \text{ mol}$

- ✓ $n(\text{H atoms}) = 2 (6.67 \times 10^{-3}) = 1.33 \times 10^{-2} \text{ mol}$
- ✓ $n(\text{C}) = \frac{0.600}{90} = 6.67 \times 10^{-3} \text{ mol}$
- ✓ $n(\text{C}) : n(\text{H atoms}) = 6.67 \times 10^{-3} : 1.33 \times 10^{-2} = 1 : 2$
- ✓ since each $-\text{OH}$ group produces one H atom, there are two $-\text{OH}$ groups.
- ✓ since C undergoes condensation with 2,4-DNPH $\Rightarrow$ aldehyde / ketone present
- ✓ since C undergoes (mild) oxidation with Fehling's $\Rightarrow$ aldehyde present

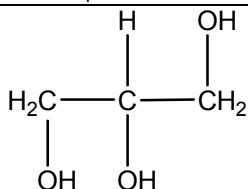
structure of C:



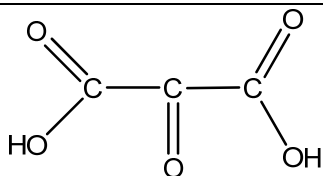
[7]

(d) Compound **C** can be both oxidised and reduced.

- (i) Give the structural formula of the compound formed when **C** is reacted with NaBH_4 under suitable conditions.



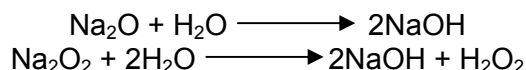
- (ii) Give the structural formula of the compound formed when **C** is heated under reflux with acidified $\text{K}_2\text{Cr}_2\text{O}_7$.



[2]

[Total: 20]

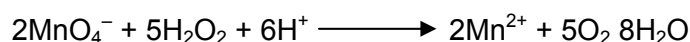
- 7 (a) When sodium is burned in air, a mixture of sodium oxide, Na_2O , and sodium peroxide Na_2O_2 , is formed. The mixture reacts with water according to the following equations.



The following information will allow you to calculate the relative amounts of the two oxides produced when sodium is burned.

- The mixture obtained by burning a sample of sodium was dissolved in distilled water and made up to 100 cm^3 to give solution **D**.
- A 25.0 cm^3 portion of solution **D** was titrated with $0.100\text{ mol dm}^{-3}\text{ HCl}$. 22.5 cm^3 of acid was required to reach the end-point.
- The H_2O_2 content of solution **D** was found by titration of another 25.0 cm^3 portion with $0.0200\text{ mol dm}^{-3}\text{ KMnO}_4$.

The following reaction occurs.



10.0 cm^3 of KMnO_4 solution was required to reach the end-point.

- (i) Using the results of the HCl titration calculate the total amount in moles of NaOH in 100 cm^3 of solution **D**.

(i) No. of moles of $\text{H}^+ = \frac{22.5}{1000} \times 0.100 = 2.25 \times 10^{-3}\text{ mol}$
 $\therefore$ No. of moles of OH^- in 25.0 cm^3 of solution **D**
 $= 2.25 \times 10^{-3}\text{ mol}$
 $\therefore$ No. of moles of OH^- in 100 cm^3 of solution **D**
 $= 2.25 \times 10^{-3} \times 4$
 $= \underline{9.00 \times 10^{-3}\text{ mol}}$ (3sf)

- (ii) Using the results of the KMnO_4 titration, calculate the amount in moles of H_2O_2 in 100 cm^3 of solution **D**.

(ii) No. of moles of $\text{KMnO}_4 = \frac{10.0}{1000} \times 0.0200 = 2.00 \times 10^{-4}\text{ mol}$
 $\therefore$ No. of moles of H_2O_2 in 25.0 cm^3 of solution **D**
 $= 2.00 \times 10^{-4} \times \frac{5}{2}$
 $= 5.00 \times 10^{-4}\text{ mol}$
 $\therefore$ No. of moles of H_2O_2 in 100 cm^3 of solution **D**
 $= 5.00 \times 10^{-4} \times 4$
 $= \underline{2.00 \times 10^{-3}\text{ mol}}$ (3sf)

- (iii) Hence calculate the amount in moles of Na_2O_2 formed during the burning of the sodium sample.

(iii) In 100 cm^3 of solution D,
 No. of moles of $\text{Na}_2\text{O}_2 = \text{No. of moles of } \text{H}_2\text{O}_2$
 $= \underline{2.00 \times 10^{-3} \text{ mol}}$ (3sf)

- (iv) Using your result in (a)(iii), calculate the amount in moles of Na_2O formed during the burning of the sodium sample.

No. of moles of NaOH from dissolving Na_2O_2 in water
 $= 2.00 \times 10^{-3} \times 2$
 $= 4.00 \times 10^{-3} \text{ mol}$

No. of moles of NaOH from dissolving Na_2O in water
 $= 9.00 \times 10^{-3} - 4.00 \times 10^{-3}$
 $= 5.00 \times 10^{-3}$

No. of moles of $\text{Na}_2\text{O} = 5.00 \times 10^{-3} \div 2 = \underline{2.50 \times 10^{-3} \text{ mol}}$ (3sf)

[6]

- (b) A buffer solution can be prepared by mixing 500 cm^3 of 4 mol dm^{-3} of CH_3COOH in 500 cm^3 of 0.5 mol dm^{-3} of NaOH.

- (i) Calculate the pH of NaOH before mixing is carried out.

$\text{pOH} = -\log [\text{OH}^-]$
 $= -\log(0.5)$
 $= 0.301$

$\text{pH} = 14 - \text{pOH}$
 $= 14 - 0.301$
 $= 13.7$

- (ii) By using suitable equations, describe how the above buffer solution works.

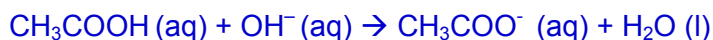
[4]

When a small amount of acid, H^+ ions, is added to the buffer solution,



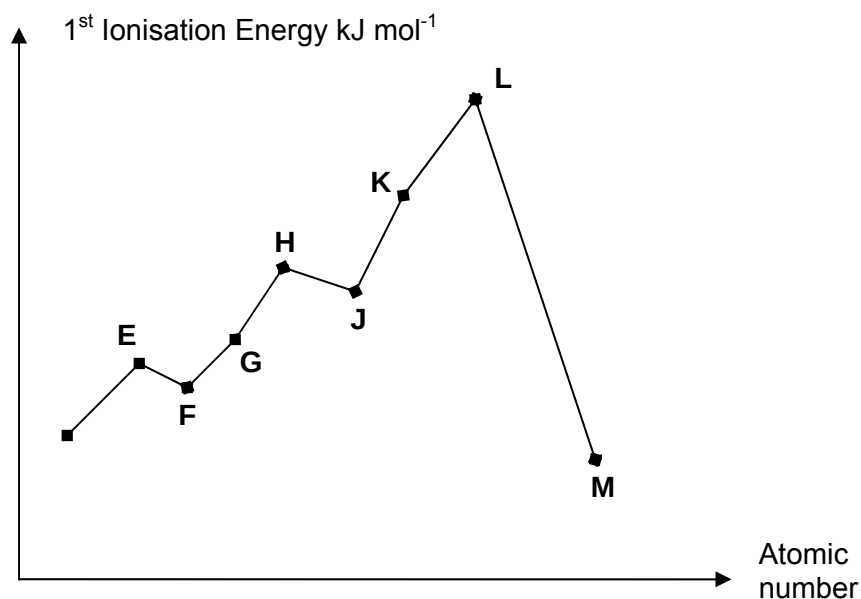
The additional H^+ ions are removed by the large concentration of CH_3COO^- present in the buffer. Hence, the pH remains almost unchanged.

When a small amount of base, OH^- ions, is added to the buffer solution,



The additional OH^- ions are removed by the large concentration of CH_3COOH present in the buffer. Hence, the pH again remains almost unchanged.

- (c) The graph below shows the first ionisation energy of eight elements with consecutive proton number.



(i) State the valence electronic configuration of element F.

ns^2np^1

(ii) Give the formula of the oxide that F is likely to form.

F_2O_3

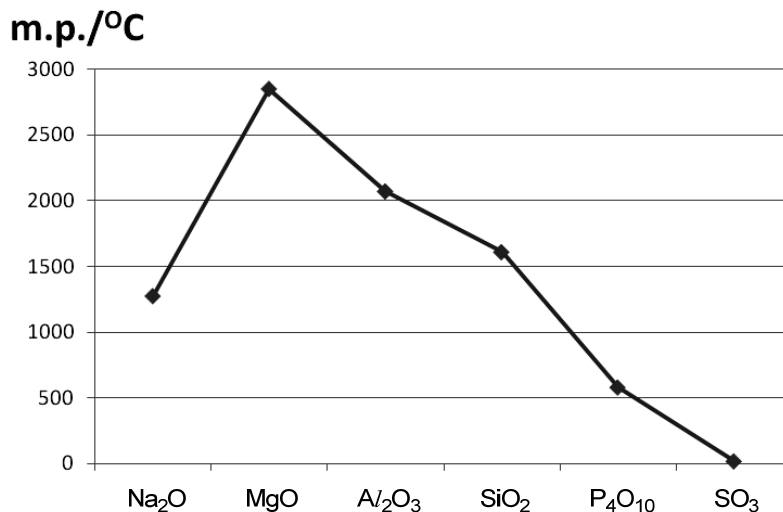
(iii) Hence or otherwise, explain the drop in first ionisation energy from E to F.

E: ns^2
F: ns^2np^1

For F, the outermost electron is removed from the (n)p subshell, which is further away from nucleus than (n)s subshell in E. Hence the outermost electron in F is less strongly attracted by the nucleus and requires less energy to remove.

[4]

(d) (i) Sketch a graph of the **melting point of oxides** of elements from sodium to sulfur.



(ii) Explain as fully as you can why the melting point varies in the way shown.

Na_2O , MgO and Al_2O_3 are **giant ionic structure** with **strong electrostatic forces of attraction between the ions**. **High amount of energy** needed to break the strong electrostatic attraction.

MgO has a higher melting point than Na_2O because of the **stronger electrostatic attraction between the highly charged Mg^{2+} and the O^{2-} ions**.

Al_2O_3 has a slightly lower melting point than MgO because Al_2O_3 has some degree of **covalent character due to the highly charged density of Al^{3+} ion**.

SiO_2 has a **giant covalent structure** with **strong extensive covalent bond between Si and O atom**. **High amount of energy** needed to break the strong extensive covalent bonds between the Si and O atoms

Both P_4O_{10} and SO_3 has a **simple molecular structure** with **weak van der Waals' forces of attraction between the molecules**. **P_4O_{10} has a larger electron cloud as compared to SO_3** . **More energy** is needed to overcome the forces of attraction between P_4O_{10} molecules leading to a higher melting point as compared to SO_3 .

[6]

[Total: 20]

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